

HE Tiancheng

Research Interests

My research interests are in probability theory and statistical physics, especially critical and near-critical phenomena in planar percolation and random-cluster models.

Education

Université de Genève Ph.D. in Mathematics Advisor: Prof. Hugo Duminil-Copin	Geneva, Switzerland Sep. 2023–Jun. 2027 (expected) Probability and statistical physics
École Normale Supérieure “Diplôme de l’ENS”, Mathematics – Thesis: “Scaling relations in the 2D near-critical planar random cluster model”	Paris, France Sep. 2020–Sep. 2023
Université Paris-Saclay Master 2 “Mathématiques de l’aléatoire” (Mathematics of randomness) – Score: 17.59/20, mention “Très Bien” (Highest mention) – Thesis: “Geometric properties of some models in Brownian geometry”	Orsay, France Sep. 2021–Sep. 2022
Peking University B.S. in Mathematics and Applied Mathematics – Total GPA: 3.72/4.00, Core GPA: 3.82/4.00 – Thesis: “Brownian disk and Brownian sphere”	Beijing, China Sep. 2017–Sep. 2022

Publications

1. H.-B. Chen, H. Duminil-Copin, **T. He**, F. Jacopin, D. Krachun, I. Manolescu and J. Xia. “Critical exponents for planar random-cluster model with cluster-weight $q = 4$.” *arXiv preprint*, arXiv:2605.30030, 2026.

Academic Service

Organizer , Geneva Math-Physics Seminar <i>Université de Genève</i>	2024–Present
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Teaching

Teaching Assistant at Université de Genève, <i>Analysis I</i>	Spring 2026
Teaching Assistant at Université de Genève, <i>Analysis I</i>	Fall 2025
Teaching Assistant at Université de Genève, <i>Mathematical theory of percolation</i>	Spring 2025
Teaching Assistant at Université de Genève, <i>General Mathematics</i>	Fall 2024

Awards

Gold Medal in International Mathematical Olympiad (58th IMO, Rio de Janeiro, Brazil)	2017
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Languages

- **English:** Fluent
- **Chinese:** Native
- **French:** Intermediate (B2)